

CD44 promoter-reporter architectures

5' -> 3' display, genomic 35133721 -> 35233402 | strand + | assembly GRCh38 | annotation Ensembl live GRCh38; selected reporter transcripts | upstream 5000 bp | downstream 1000 bp

Canonical reporter comparison: 2 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.



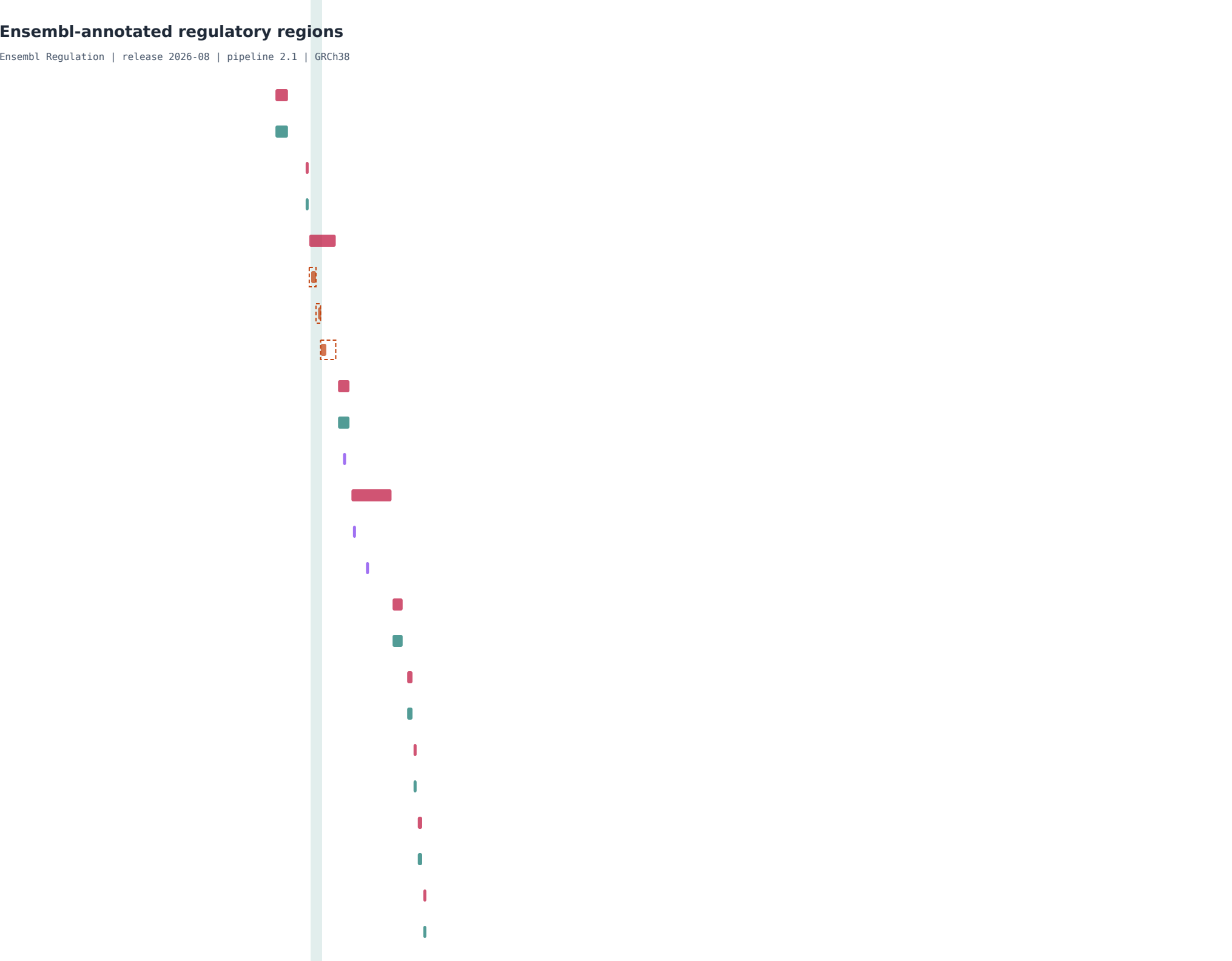
TSS stretches: reference for similarity below



Proposed reporter architectures



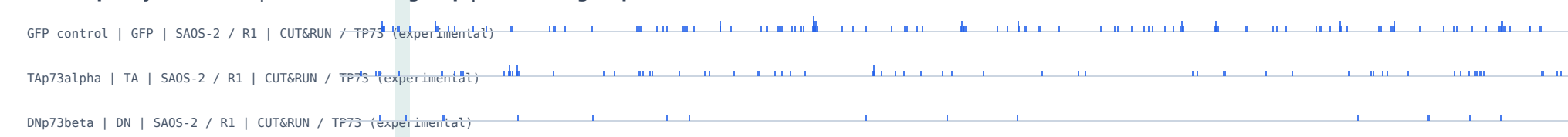
Ensembl-annotated regulatory regions



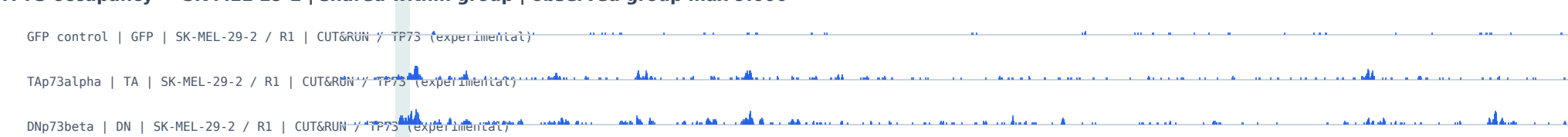
64 additional source-bound row(s) omitted from SVG; complete rows remain in JSON.

Projected occupancy grouped by declared biological context

TP73 occupancy — SAOS-2 | shared within group | observed group max 3.000



TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 9.000



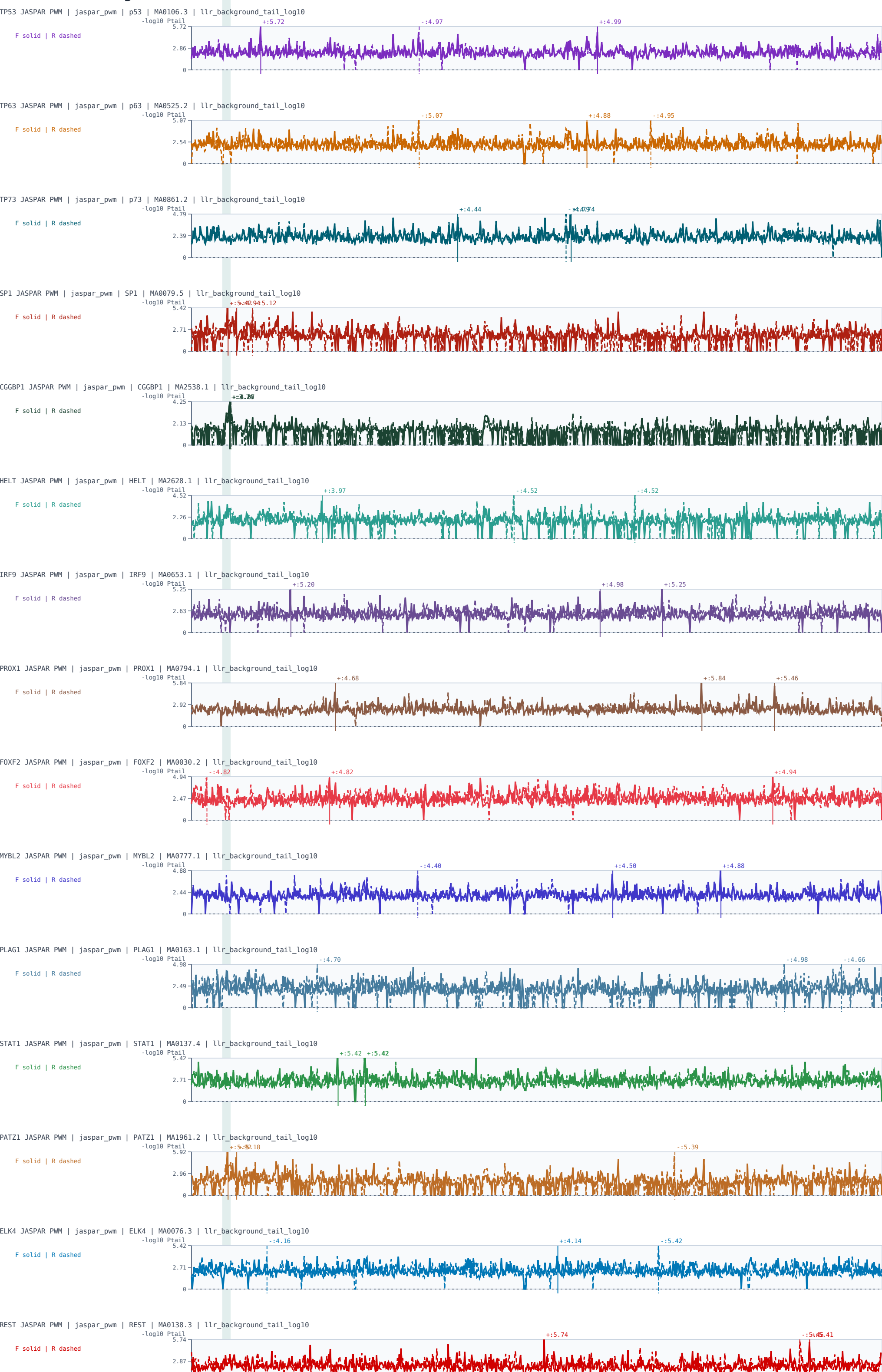
H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 6.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 44.000



Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript-oriented -500/+200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, EslE-05; self-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.



Similarity is structural evidence; Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation details

Transcript annotation alone does not establish which TSS is used in the assayed cells.

CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.

Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

Evidence provenance

ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: CD44_panel
occupancy_layout: tp73_saos2_tp73_skmel29_2_h3k4me3_saos2_h3k4me3_skmel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:GFP:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:GFP:R1